

IMMANUEL RAYNALDO SANTJOKO

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A passionate tinkerer and engineer who loves to design, build, and play with robots. Experienced in the design and development of robotics, industrial automation, and embedded systems. Dedicated to pushing the boundaries of technology through theoretical examination, hands-on experimentation, and innovation.

EDUCATION

PhD of Electrical Engineering
University of Houston, USA.

Aug. 2025 - Present

Bachelor of Engineering in Electrical (Mechatronics) Engineering
Parahyangan Catholic University, Indonesia. GPA : 3.85/4.00

Sep. 2020 - Feb. 2025

PUBLICATION

T. A. Tamba, I. R. Santjoko, Y. Y. Nazaruddin and V. Nadhira, "A multi-UAV coordination scheme for tracking control of multiple moving target objects," *Proc. IEEE International Conference on Smart Mechatronics*, Yogyakarta, Indonesia, 2024, pp. 7-11. DOI: 10.1109/ICSMech62936.2024.10812277

I. R. Santjoko, T. A. Tamba and A. Sadiyoko, "Modeling and cascade PID controller design of a spinbath circulation process," *Proc. FORTEI-International Conference on Electrical Engineering*, Badung, Indonesia, 2024, pp. 183-188. DOI: 10.1109/FORTEI-ICEE64706.2024.10824351

RECOGNITION

Krisyadana Scholarship for Academic Achievement, Parahyangan Catholic University (2022 - 2024)
Best Graduate of Mechatronics Department, Parahyangan Catholic University (Feb. 2025)

WORK EXPERIENCE

Research Assistant
University of Houston, USA.

Aug 2025 - Present

- LLM Robots

Research Assistant
Parahyangan Catholic University, Indonesia.

Sep 2024 - July 2025

- Experimental design of safety-preserving controller for the prototype of a six degrees-of freedom (6-DOF) Stewart robotic platform
- Design and analysis of cooperative simultaneous localization and pursuit (SLAP) algorithm for multi-UAV cooperative control design with application in tracking and pursuit of moving unmanned surface vehicles
- Modeling and cascade controller design for a spinbath circulation process control to optimize the performance of the overall process

Teaching Assistant
Parahyangan Catholic University, Indonesia.
Assisting Engineering Mathematics and Automatic Control Systems courses

Sep 2024 - Jul 2025

Laboratory Assistant
Parahyangan Catholic University, Indonesia.

Sep 2022 - Jul 2023, Sep 2024 - Jan 2025

Develop modules and supervise Mechatronics laboratory experiments for such courses as Electronic Systems, Embedded and Microprocessors Systems, and Automatic Control Systems

Automation Engineer Intern

Feb 2024 - Jun 2024

Asia Pacific Rayon, Indonesia.

Learned how to design a decentralized control system (DCS), perform maintenance, analyze and resolve fluctuation issues in the spinbath liquid flow, and conducting research on Spinbath Circulation Automation. Made a simulation of cascade PID control system to automate spinbath circulation process based on identification system model.

ORGANIZATION EXPERIENCES

MAHITALA (Hiker and Nature Lover Student Club)

Chief Organizer of the Diving Expedition, Bokan Kepulauan, Indonesia

May 2023 - Sep 2023

Disaster Response Volunteer, Cianjur, Indonesia

Nov 2022

HMPSTEM (Mechatronics Engineering Student Association)

Secretary and Treasurer

Jan 2023 - Des 2023

DEVELOPMENT PROJECTS

Stewart Platform, a 6-DOF parallel manipulator

2025 - Present

Cooperative SLAP, a multi-tracker system to track & pursuit moving targets using range sensor

2024 - Present

Spinbath Circulation Automation, a simulation of cascade PID control system to automate spinbath circulation process based on identification system model

2024

Mecanum Robot, a mobile robot that uses a combination of directional forces to achieve omnidirectional movement

2023

FPV Drone, a remotely piloted aircraft system that provides a real-time video feed from an onboard camera to the pilot

2022

Sumo Robot, a remotely controlled combat robot designed to compete in sumo-style robot competitions

2021

Electrical Wheelchair, a device designed to assist individuals with limited mobility

2020

REFERENCE

Tua Agustinus Tamba, Ph.D.

Associate Professor in Control & Optimization

Department of Electrical Engineering, Parahyangan Catholic University

Jl. Ciumbuleuit no. 94, Bandung 40141, West Java, Indonesia

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